

Email Security and PGP



UNIT V

SECURITY PRACTICE AND SYSTEM SECURITY Electronic Mail security – PGP, S/MIME – IP security – Web Security – SYSTEM SECURITY: Intruders – Malicious software – viruses – Firewalls.

UNIT V – SECURITY PRACTICE AND SYSTEM SECURITY

1. Electronic Mail Security

Electronic mail security protects e-mails from unauthorized access, modification, and attacks.

Main goals:

- Confidentiality
- Authentication
- Integrity
- Non-repudiation

Methods used:

- Encryption
- Digital signatures
- Hash functions

Two important e-mail security standards:

1. PGP
2. S/MIME

2. PGP (Pretty Good Privacy)

Pretty Good Privacy

PGP is an e-mail security system developed by Phil Zimmermann for secure communication.

Services Provided by PGP

- Authentication
- Confidentiality
- Compression

- E-mail compatibility
 - Segmentation
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Working of PGP

Step 1: Message Creation

Sender writes the message.

Step 2: Hashing

Hash function creates a message digest.

Step 3: Digital Signature

Digest is encrypted using sender's private key.

Step 4: Compression

Message is compressed.

Step 5: Encryption

Compressed message is encrypted using symmetric key.

Step 6: Key Encryption

Symmetric key is encrypted using receiver's public key.

Step 7: Transmission

Encrypted message is sent.

Advantages of PGP

- Strong security
- Efficient
- Provides authentication and confidentiality
- Widely used

Disadvantages

- Complex key management
- Difficult for beginners

3. S/MIME (Secure/Multipurpose Internet Mail Extensions)

S/MIME

S/MIME is a standard used to secure e-mail communication using public key cryptography and MIME.

Features

- Encryption
- Digital signatures
- Integrity
- Authentication

Working of S/MIME

1. Sender creates message.
2. Message digest is generated.
3. Digest encrypted with sender's private key.
4. Message encrypted with session key.
5. Session key encrypted with receiver's public key.
6. Receiver decrypts and verifies.

Advantages

- Standardized
- Supported by many e-mail applications
- Strong authentication

Disadvantages

- Requires certificates
- Certificate management is difficult

Difference Between PGP and S/MIME

PGP	S/MIME
Uses web of trust	Uses CA certificates

PGP	S/MIME
Individual-based	Organization-based
More flexible	More standardized
Free versions available	Common in enterprises

4. IP Security (IPSec)

IPsec

IPSec is a set of protocols used to secure IP communication.

Services of IPSec

- Authentication
- Confidentiality
- Data integrity
- Access control

Components of IPSec

1. Authentication Header (AH)

Provides:

- Authentication
- Integrity

But does not provide encryption.

2. Encapsulating Security Payload (ESP)

Provides:

- Encryption
- Authentication
- Integrity

Modes of IPSec

Transport Mode

- Encrypts only data part.
- Header remains unchanged.

Tunnel Mode

- Encrypts entire IP packet.
 - Used in VPNs.
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Advantages

- Strong network security
- Transparent to applications
- VPN support

Disadvantages

- Complex configuration
 - Processing overhead
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5. Web Security

Web security protects websites and web applications from attacks.

Common Web Threats

- Phishing
 - SQL Injection
 - Cross-site scripting (XSS)
 - Session hijacking
 - Malware
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Web Security Mechanisms

SSL/TLS

Transport Layer Security

Provides secure communication between browser and server.

Features:

- Encryption
- Authentication
- Integrity

HTTPS uses SSL/TLS.

Cookies

Small data stored in browser for session management.

Secure Socket Layer (SSL)

Older protocol used for secure communication.

6. Intruders

An intruder is an unauthorized person accessing a system.

Types of Intruders

1. Masquerader

External person pretending to be authorized user.

2. Misfeasor

Authorized user accessing unauthorized data.

3. Clandestine User

User who gains supervisory control and hides activities.

Intrusion Detection Techniques

- Password protection
 - Firewalls
 - IDS (Intrusion Detection System)
 - Log monitoring
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7. Malicious Software (Malware)

Malware is software designed to damage systems or steal data.

Types of Malware

1. Virus

Attaches to files and spreads.

2. Worm

Self-replicating malware spreading through networks.

3. Trojan Horse

Appears legitimate but performs malicious actions.

4. Spyware

Collects user information secretly.

5. Ransomware

Locks files and demands payment.

8. Viruses

A virus is a malicious program that replicates itself.

Phases of Virus

1. Dormant Phase

Inactive stage.

2. Propagation Phase

Virus spreads.

3. Triggering Phase

Activated by event.

4. Execution Phase

Performs malicious action.

Types of Viruses

- Boot sector virus
 - File infector virus
 - Macro virus
 - Multipartite virus
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Prevention Methods

- Antivirus software
 - Regular updates
 - Avoid suspicious files
 - Backup data
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9. Firewalls

Firewall

A firewall monitors and controls incoming and outgoing network traffic.

Functions

- Prevent unauthorized access
 - Filter packets
 - Monitor traffic
 - Protect internal network
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Types of Firewalls

1. Packet Filtering Firewall

Filters packets based on rules.

2. Stateful Inspection Firewall

Tracks active connections.

3. Application-Level Gateway

Filters application-specific traffic.

4. Proxy Firewall

Acts as intermediary between user and internet.

Advantages

- Network protection
- Prevents attacks
- Access control

Disadvantages

- Cannot stop all attacks
 - May reduce performance
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Short Notes

PGP

E-mail security system providing encryption and digital signature.

S/MIME

Secure e-mail standard using certificates.

IPSec

Protocol suite securing IP communication.

Firewall

Security system filtering network traffic.

Virus

Self-replicating malicious software.

Intruder

Unauthorized user accessing system.